Research Task Assignment

Advancements in Hardware for Mixed Reality

Assignment Details

Work Mode: Individual or pairs

Presentation Date: Monday, February 23, 2026

Presentation Duration: 10 minutes

Topic: Advancements in Hardware for Mixed Reality (MR)

Research Focus

Your research should explore recent technological innovations in MR hardware, including display technologies, optical systems, sensors, tracking systems, processing units, and wearable device form factors. Consider advancements in microdisplays, eye tracking, hand tracking, spatial computing, and the integration of AI with MR hardware.

Suggested Research Papers

Below are 20 recent research papers on MR hardware advancements. These papers cover various aspects including display optics, sensors, interaction techniques, spatial computing. Please note you can select your own research (send me an email for approval). The form will be available for topic selection tomorrow morning (17th of feb 20206) and will be closed on wednesday.

1-Advancements and Innovations in Mixed Reality: A Comprehensive Review

2-Advances and challenges in microdisplays and imaging optics for virtual reality and mixed reality

3- Mixed Reality with Hardware Acceleration: implementing the multimodal user interface vision

4- Creating a Mixed Reality Retail Experience With Present Hardware Limitations

5- Mixed reality virtual device (MRVD) for seamless MR-IoT-digital twin convergence

6- Exploring mobile devices as haptic interfaces for mixed reality

7- Advances and challenges in microdisplays and imaging optics for virtual reality and mixed reality

8- Video see-through mixed reality with focus cues

9- Predicting hand-object interaction for improved haptic feedback in mixed reality

10-Novel challenges of safety, security and privacy in extended reality

11- Enhanced human–robot interface with operator physiological parameters monitoring and 3d mixed reality

12- Mixed reality tunneling effects for stereoscopic untethered video-see-through head-mounted displays

13- Latency and sampling compensation in mixed-reality-in-the-loop simulations of production systems

14- Real-time mixed reality display of dual particle radiation detector data

Additional Research Resources

Google Scholar: Search for specific topics like “mixed reality displays,” “eye tracking MR,” “spatial computing hardware”

IEEE Xplore: Database of IEEE conference and journal papers on XR technologies

ArXiv.org: Preprints of latest research in computer vision, graphics, and human-computer interaction

ACM Digital Library: Papers from CHI, UIST, SIGGRAPH conferences on XR interaction and displays

Presentation Guidelines

• Focus on 2-3 major hardware innovations in depth rather than covering everything superficially

• Include diagrams or images of hardware components and systems

• Explain the technical principles behind the advancements

• Discuss practical applications and future implications

• Compare different approaches or technologies where relevant

• Prepare for Q&A by understanding the fundamentals deeply

Good luck with your research!